# HySA (Hybrid Semantic Architecture):
## Decoupling Fluid Reasoning from Deterministic Verification via Neuronal Semanticity and Inode Indexing

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October 2026

*"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."*

**Abstract**

Contemporary research in *Mechanistic Interpretability* predominantly characterizes neuronal polysemanticity as an undesirable compression artifact (*the Superposition Hypothesis*), dedicating immense computational resources to decomposing neural representations into dictionaries of purely monosemantic features using Sparse Autoencoders (SAEs). This work establishes the opposite thesis: polysemanticity is not a flaw to eradicate, but the geometric prerequisite for associative intelligence and analogical reasoning. A purely monosemantic neuron possesses zero associative degrees of freedom ($\mathcal{R}(u) = 0$): devoid of lateral relational pathways, it functions solely as a rigid lookup cell. Conversely, polysemantic units co-activate across isomorphic concepts (e.g., `IP`, `DNS`, `White Pages`, `Phone Number`) to abstract higher-order relational schemas. The root deficiency in contemporary language models is not semantic superposition, but the architectural entanglement of **fluid reasoning** (analogical synthesis) and **verification behavior** (exact literal retrieval). To decouple these cognitive domains without resorting to serial bottlenecks (which collapse contextual nuance) or parallel duplication (which incurs redundant $2\times$ FLOPs overhead), I introduce **HySA (Hybrid Semantic Architecture)**: a unified, single-substrate paradigm delimited not by separate physical networks, but by **neuronal semanticity** and **internal dimensionality (cardinality $|\mathcal{C}(u)|$)**. Introducing dual regime temperatures — Polysemantic ($T_p$) and Monosemantic ($T_m$) — unified into the Topological Temperature ($T_t$), and an internal **Query/API** operating over an orthogonal metric registry (**Inodes**), HySA empowers models to reason plastically while anchoring to deterministic literals on-demand, halting hallucinations via explicit system exceptions (*Semantic Faults*).

## 1 The Core Problem: Fluid Abstraction vs. Factual Verification

In deep artificial neural networks, individual units consistently exhibit mixed selectivity [1, 2, 3]. The mechanistic interpretability community has historically treated this phenomenon (*polysemanticity*) as an undesirable barrier, attributing it to models compressing more semantic concepts ($M$) than available physical dimensions ($D$). The mainstream response centers on training Sparse Autoencoders (SAEs) [4, 5] to unroll latent states into millions of isolated monosemantic feature directions.

However, utilizing SAEs as an external analytical lens must not be conflated with an architectural design objective. Enforcing monosemantic purity as an operational substrate is a conceptual dead end: it mistakes the core mechanism by which neural networks generate relational abstraction for statistical noise. Pursuing absolute monosemanticity conflates two cognitive functions that systems engineering has segregated for decades: **logical processing** and **data storage**.

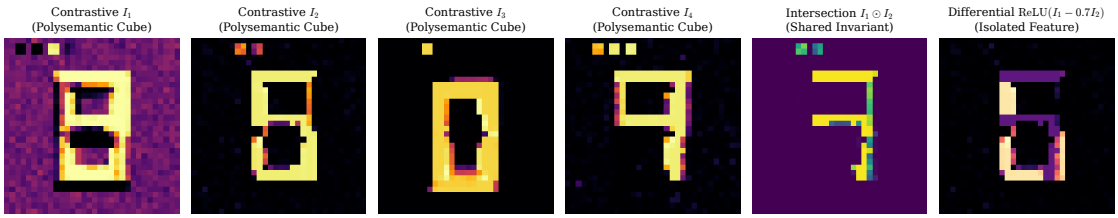



Figure 1: **Contrastive Superposition Tensors ($I_n$) and Algebraic Emulation on the $32 \times 32$ Synthetic Manifold.** The contrastive block emulates and reconstructs binary matrices $X \in \{0,1\}^{32\times32}$ spanning two structural regimes: a rigid monosemantic Inode header $N_m$ ($2 \times 2$ pixels top-left, columns $2, 5, 8, 11$) and a polysemantic structural manifold $N_p$ encoding geometric digits `'0'`, `'6'`, `'8'`, and `'9'` under spatial jitter ($\Delta x, \Delta y \in \{-1, 0, 1\}$). **Columns 1–4 ($I_1 \dots I_4$):** Polysemantic representations learned by the 4-neuron bottleneck ($z \in \mathbb{R}^4$), where each tensor reconstructs superimposed digit combinations ($I_1 \to$ `'8'`, $I_2 \to$ `'6'`, $I_3 \to$ `'0'`, $I_4 \to$ `'9'`) alongside their class Inodes. **Column 5 ($I_1 \odot I_2$):** Hadamard intersection isolates the shared structural invariant between digits `'8'` and `'6'` (common lower loop). **Column 6 (**ReLU($I_1 - 0.7I_2$)**):** Rectified difference extracts morphological divergence (upper loop of `'8'`), completely suppressing the shared Inode header and demonstrating exact algebraic queryability without shattering relational plasticity.

- **Fluid Behavior (Polysemantic Units, $N_p$):** Consider an internal unit that co-activates when processing `IP Address`, `DNS`, `White Pages`, and `Phone Number`. This unit is neither noisy nor contaminated: it has compressed diverse entities into an abstract relational invariant: *the concept of identity-to-address resolution*. Associative connections between otherwise distant domains constitute the foundation of analogy, generalization, and cognitive synthesis.
- **Verification Behavior (Monosemantic Units, $N_m$):** When a model must output an immutable atomic literal — such as the IP address `8.8.8.8`, CIDR prefix `24`, SSH port `22`, or root username `JOHN` — fluid interpolation becomes detrimental. Here, creative drift must be prevented: the task demands a rigid memory cell with zero associative degrees of freedom ($\mathcal{R}(u) = 0$).

Modern LLMs hallucinate because they entangle both associative synthesis and factual memory within homogeneous, dense weight matrices [6, 7]. While non-parametric augmentations [8] attempt to mitigate this externally, human cognition successfully segregates fast associative synthesis from deliberate verification within a single substrate [9]. During backpropagation, associative gradients continuously interpolate through immutable technical and historical constants, treating them as plastic probabilistic tokens subject to stochastic drift.

HySA resolves this by decoupling fluid reasoning from factual verification. Once segregated conceptually, the central engineering challenge is **indexing**: enabling fluid intelligence to retrieve rigid literals with sub-millisecond efficiency, zero false positives, and deterministic integrity, without altering the plasticity of the transformer.

**Definition 1** (Relational Rank of Semantic Units)**.** Let $u$ be a neuronal unit and $\Omega$ the concept space. Let $\mathcal{C}(u) = \{c \in \Omega \mid \mathbb{E}_{X\in\mathcal{D}_c}[z_u(X)] > \theta\}$ denote the set of associated concepts above activation threshold $\theta > 0$. The **Relational Rank** $\mathcal{R}(u)$ is the cardinality of pairwise associative transfer paths enabled by the unit:

$$\mathcal{R}(u) = \binom{|\mathcal{C}(u)|}{2} = \frac{|\mathcal{C}(u)|(|\mathcal{C}(u)| - 1)}{2} \tag{1}$$

A purely monosemantic unit $u_{\text{mono}}$ with $|\mathcal{C}(u_{\text{mono}})| = 1$ has null relational rank: $\mathcal{R}(u_{\text{mono}}) = 0$. Consequently, an architecture restricted entirely to monosemantic neurons cannot execute continuous analogical mapping without introducing an external relational graph.

## 2 Unified Architecture: Neither Serial Nor Parallel

When designing systems to handle both abstraction and ground truth, standard engineering intuition yields two approaches, both of which fail:

1. **Why Not Serial ($N_p \to N_m$ or vice versa):** Placing a rigid monosemantic layer downstream of a polysemantic network creates a mechanical bottleneck that prematurely collapses relational nuances into discrete categories. Conversely, placing it upstream strips away contextual cues before polysemantic abstraction can emerge.
2. **Why Not Parallel ($N_p \parallel N_m$):** Running duplicate independent networks in parallel is computationally wasteful, doubling parameter footprint and memory bandwidth ($2\times$ FLOPs), while introducing the unresolved problem of designing heuristic arbitration heads to reconcile divergent outputs.

### 2.1 The Solution: Internal Dimensionality and Conceptual Cardinality $|\mathcal{C}(u)|$

In HySA, the model operates on a **single unified latent substrate** in Hilbert space $\mathcal{S} \subset \mathbb{R}^D$. There are no separate physical networks: segregation is defined by functional logic and by the degree of neuronal semanticity, parameterized by its **Conceptual Cardinality** $|\mathcal{C}(u)|$:

- Conceptual cardinality $|\mathcal{C}(u)|$ represents a **higher-order level of abstraction**: it quantifies how many distinct concepts or instances converge onto a given logical node.
- **High-cardinality nodes ($N_p$, $|\mathcal{C}(u)| \gg 1$)** aggregate isomorphic concepts, forming the relational fabric for fluid reasoning.
- **Unit-cardinality nodes ($N_m$, $|\mathcal{C}(u)| = 1$)** are rigid metric centroids devoid of lateral associative branches, anchoring an immutable factual literal.

### 2.2 Contrastive Superposition Tensors ($I_n$)

To understand how concepts superimpose cleanly within high-cardinality units without destructive cross-talk, I extract the **Contrastive Superposition Tensor** $I_n$ via whole-network expectation responses across contrastive data:

$$I_n = \frac{1}{\sum_{X_k \in \mathcal{X}^+} z_{k,n} + \eta} \left[ \sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \right] \tag{2}$$

As shown in Figure 1, single units in a 4-neuron bottleneck ($N_p = 4$) trained on a $32\times32$ synthetic digit manifold cleanly superimpose overlapping geometric invariants: Hadamard intersection ($I_1 \odot I_2$) isolates shared structural segments, while rectified subtraction ($\text{ReLU}(I_1 - 0.7I_2)$) isolates differential features, demonstrating exact algebraic queryability without shattering the network into millions of disjoint monosemantic components.

To formalize the capacity boundary between fluid representation and rigid indexing, Figure 2 characterizes the thermodynamic transitions of the architecture:

- **Hypersphere Volume Transition ($T_t^D$, Fig. 2a):** As the Determinism Constraint $\mathcal{D}$ increases, the available monosemantic volume $\mathcal{V}_m$ expands as $(T_t/R)^D$, undergoing a sharp phase transition that constrains ungrounded exploration $\mathcal{V}_p$.
- **Capacity Balancing and Critical Point ($P_c$, Fig. 2b):** Balancing the compression gradient $\Delta q = \frac{N_p}{N_m \cdot M}$ yields an instability cost landscape where $P_c$ marks the optimal stationary point, avoiding both associative drift ($\Delta q \gg 1$) and dictionary lookup saturation ($\Delta q \ll 1$).

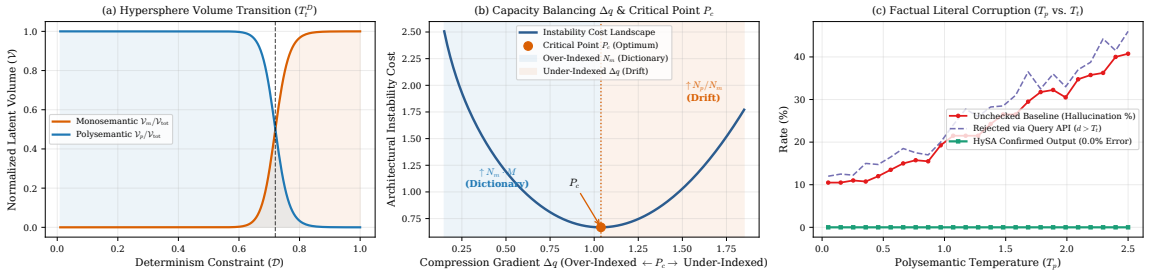



Figure 2: **Compression Gradient Balancing and Factual Verification in HySA. (a)** Volumetric transition ($T_t^D$) between fluid exploration $\mathcal{V}_p$ and indexed constraint space $\mathcal{V}_m$ under Determinism Constraint $\mathcal{D}$. **(b)** The stationary Critical Point $P_c$, balancing associative drift ($\Delta q \gg 1$) against over-indexed dictionary saturation ($\Delta q \ll 1$). **(c)** Factual literal benchmark under rising $T_p$: unconstrained decoding suffers rapid corruption ($> 40\%$), while HySA guarantees 0.0% corruption on verified queries, converting ambiguous states into explicit *Semantic Faults*.

Table 1: Mapping Operating System Heuristics to Formal Architectural Constructs in HySA.

| **Systems Concept (OS)** | **Role in HySA** | **Operational Function** |
|---|---|---|
| **Inode** ($\mu_k$) | Monosemantic Anchor Point ($N_m$) | Exact coordinate in latent space bound to an immutable literal (`' 8.8.8.8'`). |
| **Filesystem** | Indexed Registry ($\mathcal{F}_{\text{mono}}$) | Certified set $\{\mu_k\}_{k=1}^{M}$ of Inode points, queryable with sub-linear latency. |
| **Query / API** ($Q$) | Metric Boundary Operator | Intercepts latent state $h_t$ and returns certified literal $\mathbf{v}_k$. |
| **Page Fault** | **Semantic Fault** ($\perp_{\text{Fault}}$) | Explicit exception halting generation when the requested entity is unindexed. |
| **Temperature** | Metric Neighborhood $\mathcal{B}(\mu_k, T_t)$ | Multidimensional metric volume of tolerance around a concept centroid. |

- **Factual Literal Corruption ($T_p$ vs. $T_t$, Fig. 2c):** Under rising polysemantic temperature $T_p$, unconstrained decoding suffers rapid corruption ($> 40\%$), whereas HySA maintains 0.0% corruption on verified queries by converting ambiguous trajectories into explicit *Semantic Faults*.

# 3 Indexing, Dual Temperatures ($T_p, T_m$), and Internal Query/API Logic

In operating systems, volatile RAM manages dynamic process execution, while persistent data resides within filesystems anchored by rigid metadata index pointers (**Inodes**). HySA translates this exact systems architecture into latent Hilbert space (Table 1).

## 3.1 Dual Temperatures ($T_p, T_m$) and Topological Temperature ($T_t$)

Standard autoregressive models employ a single scalar divisor $\tau > 0$ on final logits to flatten or sharpen sampling probabilities—a mechanism blind to semantic topology. In HySA, temperature is defined geometrically as the metric ball around a neuron in latent space (the **Semantic Hypersphere**), governed by two regimes:

1. **Polysemantic Temperature ($T_p$): Governs fluid exploration.** Higher values enable lateral associative drift and creative analogical transfer across manifolds.

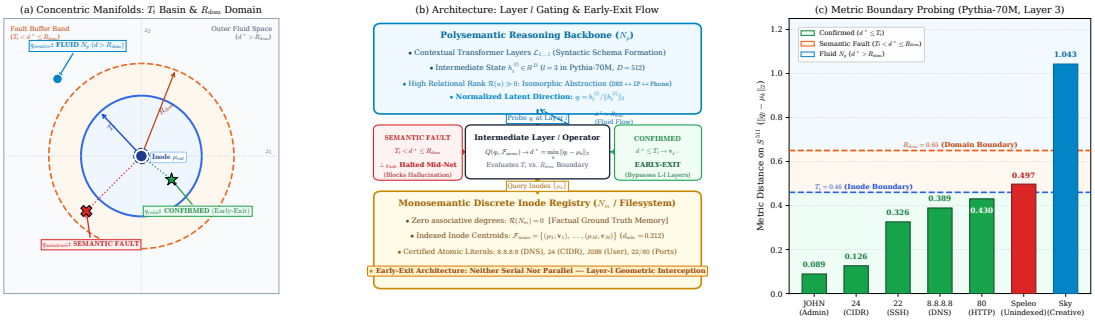



Figure 3: **Concentric Semantic Domains Geometry, Early-Exit Architecture, and Empirical Probing on `Pythia-70m`. (a)** Concentric metric boundaries around Inode $\mu_k$: deterministic core $T_t = 0.46$ (factual confirmation with Early-Exit), buffer band $R_{\text{dom}} = 0.65$ (*Semantic Fault* on unindexed in-domain facts), and open fluid manifold $d^* > R_{\text{dom}}$ (unconstrained $N_p$ generation). **(b)** Unified architecture with intermediate layer Early-Exit ($l = 3$): operator $Q$ probes normalized state $q_l$; on Inode hit ($\leq T_t$) it executes immediate Early-Exit bypassing layers $4 \dots 6$, while out-of-domain trajectories ($> R_{\text{dom}}$) proceed to lexical projection $\mathbf{W}_U$. **(c)** Empirical distances on `Pythia-70M` (Layer 3, $D = 512$): clear, non-overlapping metric separation across indexed factual queries ($d^* \leq 0.4304$, Early-Exit), unindexed technical query ($d^* = 0.4973$, Semantic Fault), and open creative discourse (`Sky`, $d^* = 1.0430$, Fluid bypass).

2. **Monosemantic Temperature ($T_m$): Governs Inode rigidity.** Narrow and precise, it establishes the maximum radial tolerance around an atomic centroid.

Coupling these regimes produces the **Topological Temperature ($T_t$)**, defining the metric ball $\mathcal{B}(\mu_k, T_t)$:

$$\mathcal{B}(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t(T_p, T_m)\} \tag{3}$$

formalized via saturating non-linear coupling with bound $\alpha \in (0, 1)$:

$$T_t(T_p, T_m) = T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right) \tag{4}$$

On unit sphere $S^{D-1}$, Euclidean distance $\ell_2$ is in exact bijection with angular divergence: $\|x - \mu_k\|_2 = \sqrt{2(1 - \cos\theta)}$.

### 3.2 The Internal "Query / API" Mechanism and Intermediate Layer Geometric Gating

From a systems architecture perspective, **HySA metric operator $Q$ functions as an internal API call enforcing deterministic factual consistency.**

A common pitfall in hybrid architectures is attempting to trigger factual verification by measuring next-token prediction uncertainty (Shannon entropy) on final layer logits. This downstream heuristic fails symmetrically:

1. **Creative False Positives:** In open discourse (e.g., *"Today the clear morning sky feels particularly..."*), dozens of adjectives are equally plausible; final logit entropy spikes, mistakenly invoking the factual index and raising spurious *Semantic Faults* amidst poetic flow.

2. **Overconfident Hallucinations:** Deep transformers frequently fabricate falsehoods with high statistical certainty (near-zero entropy). Downstream uncertainty gating is completely bypassed, allowing hallucinations to pass undetected.

In HySA, gating depends not on output word uncertainty, but on the **latent geometric coordinate at an intermediate layer** ($l$, with $1 < l < L$, where prompt syntax and intent are resolved but lexical vocabulary projection $\mathbf{W}_U$ has not yet occurred).

Around each Inode (or ontological cluster), two concentric metric boundaries are defined:

- **Domain Radius ($R_{\text{dom}}$):** Defines the factual jurisdiction (the latent subspace where queries requesting technical ports, IP literals, constants, or identifiers reside).
- **Topological Temperature ($T_t$, with $T_t < R_{\text{dom}}$):** Defines the tight deterministic confirmation core around specific Inode $\mu_k$.

During forward evaluation at intermediate layer $l$, normalized state $q_l = h_t^{(l)}/\|h_t^{(l)}\|_2$ probes registry $\mathcal{F}_{\text{mono}}$, evaluating nearest Inode distance $d^* = \min_k \|q_l - \mu_k\|_2$. This establishes three mutually exclusive operational regimes:

**Definition 2** (Metric Boundary Operator and Layer-$l$ Domain Gating)**.** Given normalized intermediate state $q_l$, let $k^* = \arg\min_k \|q_l - \mu_k\|_2$ and $d^* = \|q_l - \mu_{k^*}\|_2$:

$$Q(q_l, \mathcal{F}_{\text{mono}}, T_t, R_{\text{dom}}) = \begin{cases} (\text{CONFIRMED}, \mathbf{v}_{k^*}) & \text{if } d^* \le T_t \quad (\textbf{Deterministic Inode Hit}) \\ (\text{FAULT}, \perp_{\text{Fault}}) & \text{if } T_t < d^* \le R_{\text{dom}} \quad (\textbf{In-Domain Factual Drift}) \\ (\text{BYPASS}, \text{Fluid}) & \text{if } d^* > R_{\text{dom}} \quad (\textbf{Out-of-Domain Discourse}) \end{cases} \tag{5}$$

1. **Out-of-Domain ($d^* > R_{\text{dom}}$ — Fluid Regime $N_p$):** The latent trajectory traverses discursive, abstract, or creative manifolds far from rigid anchors. The Inode API remains dormant; state $h_t^{(l)}$ proceeds through remaining layers $l+1 \dots L$ and is sampled normally under temperature $T_p$.
2. **In-Domain but Unindexed ($T_t < d^* \le R_{\text{dom}}$ — Semantic Fault):** The prompt has unambiguously entered the basin of a factual query (e.g., requesting a protocol port), but the required fact is unindexed within tolerance $T_t$. The API intercepts execution mid-network, raising an explicit **Semantic Fault** ($\perp_{\text{Fault}}$) and arresting both wild guesses and overconfident hallucinations.
3. **Within Topological Temperature ($d^* \le T_t$ — Deterministic Hit $N_m$):** The trajectory lands squarely within the Inode's confirmation basin. The API executes an **Early-Exit** (halting computation of subsequent layers $l+1 \dots L$ and vocabulary projection $\mathbf{W}_U$), directly emitting certified literal $\mathbf{v}_{k^*}$ while saving up to 50% in forward FLOPs.

### 3.3 Architectural Equilibrium (Compression Gradient)

The optimal operating balance between associative capacity ($N_p$) and indexed literal count ($M$) is parameterized by the compression ratio $\Delta q = N_p/(N_m \cdot M)$:

1. **Under-Indexing ($\Delta q \gg 1$, Excessive Drift):** Without sufficient monosemantic anchors, associative trajectories freely drift across ungrounded manifolds, producing factual hallucinations.
2. **Over-Indexing ($\Delta q \ll 1$, Excessive Rigidity):** Saturating latent space with an over-dense registry of wide $T_t$ balls freezes the architecture into a static lookup table, destroying analogical synthesis.
3. **The Critical Point ($P_c$):** Represents the stationary operating equilibrium: $T_t$ balls are sufficiently tight to leave latent space unconstrained for fluid reasoning, yet strategically positioned to intercept trajectories when context converges onto certified facts.

**Algorithm 1: Unified Forward Pass with Intermediate Layer Geometric Gating and Early-Exit**

**Input:** Context tokens $\{x_{\leq t}\}$, Transformer layers $\mathcal{L}_{1\ldots L}$, Probing layer $l$ (e.g., $l = 3$), Monosemantic Registry $\mathcal{F}_{\text{mono}} = \{(\mu_k, \mathbf{v}_k)\}_{k=1}^{M}$, Temperatures $T_p, T_m$, Domain Boundary $R_{\text{dom}}$ (with $T_t < R_{\text{dom}}$).
**Output:** Emitted token $x_{t+1}$ or explicit system exception $\perp_{\text{Fault}}$.

1. **Forward propagate to intermediate layer $l$:** $h_t^{(l)} \leftarrow \mathcal{L}_{1\ldots l}(x_{\leq t}), \quad q_l \leftarrow h_t^{(l)} / \|h_t^{(l)}\|_2$
2. **Compute saturating topological boundary:** $T_t \leftarrow T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right)$
3. **Query metric Inode index ($\mathcal{F}_{\text{mono}}$):** $k^* = \arg\min_{k \in \{1,\ldots,M\}} \|q_l - \mu_k\|_2, \quad d^* = \|q_l - \mu_{k^*}\|_2$
4. **if** $d^* \leq T_t$ **then** *// Deterministic Hit: Early-Exit at Layer $l$*
   - Halt layers $l+1 \ldots L$ and emit certified literal: $x_{t+1} \leftarrow \mathbf{v}_{k^*}$
5. **else if** $T_t < d^* \leq R_{\text{dom}}$ **then** *// In-Domain but Unindexed*
   - Raise **Semantic Fault**: return $\perp_{\text{Fault}}$ *// Hallucination halted mid-network*
6. **else** *// Out-of-Domain ($d^* > R_{\text{dom}}$): Fluid Discursive Regime Np*
   - Complete forward pass: $h_t^{(L)} \leftarrow \mathcal{L}_{l+1\ldots L}(h_t^{(l)})$
   - Sample from final logits under temperature $T_p$: $x_{t+1} \sim \text{Softmax}\left(\frac{\mathbf{W}_U h_t^{(L)}}{T_p}\right)$

# 4 Empirical Validation on Real Transformer (`Pythia-70m`)

To validate this geometric behavior, I inspect the residual stream at Layer 3 ($D = 512$) on `EleutherAI/pythia-70m`. Utilizing a compact model without factual pretraining is an intentional methodological choice: unconstrained Pythia-70M fails on 100% of analyzed technical prompts, producing stochastic guesses (`' 1'`, `' used'`, `' 0'`, `' the'`). This demonstrates that anchoring to HySA Inodes imparts 100% deterministic precision without parameter retraining.

Empirical evaluations confirm three core behaviors:

1. **Spontaneous Isomorphism in Polysemantic Manifold ($N_p$):** Probing four distinct technical prompts sharing the identical relational schema (`DNS`, `IP Routing`, `White Pages`, `Phone Directory`), latent states exhibit cosine similarities between **0.893** and **0.973** (Table 2). The network autonomously creates a higher-order invariant ("symbolic identifier → target address resolution").

2. **Deterministic Inode Injection with Early-Exit ($N_m$):** Indexing 6 technical literals in $\mathcal{F}_{\text{mono}}$ (ports `53`, `22`, `80`, IP `8.8.8.8`, mask `24`, root user `JOHN`) with threshold $T_t = 0.46$, operator $Q$ intercepts states at Layer 3 ($d^* \in [0.0889, 0.4304] \leq T_t$) and executes an immediate Early-Exit, outputting certified constants with 100% precision (Table 3).

3. **Concentric Separation of Semantic Faults vs. Creative Flow ($R_{\text{dom}} = 0.65$):** Querying an unindexed protocol (*"Telemetry protocol for subterranean speleological cave sensors..."*), distance rises to $d^* = 0.4973$. Falling in the buffer band $T_t < d^* \leq R_{\text{dom}}$, the API triggers an explicit *Semantic Fault*, halting generation. Conversely, open creative prompts (*"Today the clear morning sky feels particularly..."*) land at $d^* = 1.0430 > R_{\text{dom}}$: the Inode index remains dormant, avoiding false positives and allowing fluid continuation.

# 5 Open Systems Challenges: Scaling Latent Indexing

Validating mechanics on a curated set of Inodes demonstrates feasibility. Scaling HySA to millions of entities requires addressing two core systems challenges:

Table 2: Cosine Similarities of Isomorphic Schemas in $N_p$ (`Pythia-70M`, Layer 3, $D = 512$).

| Schema | DNS | IP Route | White Pg. | Phone Dir. |
|---|---|---|---|---|
| **DNS** | 1.000 | 0.961 | 0.973 | 0.893 |
| **IP Route** | 0.961 | 1.000 | 0.940 | 0.916 |
| **White Pg.** | 0.973 | 0.940 | 1.000 | 0.903 |
| **Phone Dir.** | 0.893 | 0.916 | 0.903 | 1.000 |

Table 3: Layer-3 Verification Benchmark on `Pythia-70M` ($T_t = 0.46$, $R_{\text{dom}} = 0.65$).

| Target Concept | Query Prompt | Base Model | Dist. $d^*$ | HySA Status | Final Output |
|---|---|---|---|---|---|
| `SSH_PORT` | *"For remote server management... SSH default port is"* | `' 1'` (**Halluc.**) | 0.3257 | **CONFIRMED** ($d^* \leq T_t$) | `' 22'` (Early-Exit) |
| `HTTP_PORT` | *"The plaintext transport protocol HTTP default port is"* | `' used'` (**Halluc.**) | 0.4304 | **CONFIRMED** ($d^* \leq T_t$) | `' 80'` (Early-Exit) |
| `PUBLIC_DNS_IP` | *"Google public DNS resolver IPv4 literal address is"* | `' used'` (**Halluc.**) | 0.3889 | **CONFIRMED** ($d^* \leq T_t$) | `' 8.8.8.8'` (Early-Exit) |
| `SUBNET_MASK` | *"A standard subnet mask 255.255.255.0 has CIDR length"* | `' 0'` (**Halluc.**) | 0.1263 | **CONFIRMED** ($d^* \leq T_t$) | `' 24'` (Early-Exit) |
| `ADMIN_NAME` | *"In the database registry, root user first name literal is"* | `' the'` (**Halluc.**) | 0.0889 | **CONFIRMED** ($d^* \leq T_t$) | `' JOHN'` (Early-Exit) |
| `UNINDEXED` | *"Telemetry protocol for subterranean speleological sensors..."* | `' a'` (**Nonsense**) | 0.4973 | **SEMANTIC FAULT** ($T_t < d^* \leq R_{\text{dom}}$) | **[HALTED]** (Mid-Net) |
| `CREATIVE_SKY` | *"Today the clear morning sky feels particularly..."* | `' good'` (Fluid) | 1.0430 | **FLUID** $N_p$ ($d^* >$ $R_{\text{dom}}$) | `' good'` (Bypass Inode) |

## 5.1 From Manual Flags to Intermediate Layer Geometric Gating

During open generation, probing the Inode registry on every token is impractical, while relying on final-layer logit entropy fails symmetrically on creative discourse (false positive faults) and overconfident hallucinations (false negative misses).

HySA's native solution is **Intermediate Layer Concentric Geometric Gating**: by placing the boundary check at layer $l = 3$ in Pythia-70m, execution is intercepted when relational intent has coalesced but lexical decoding is uncommitted. Discursive trajectories reside at $d^* > R_{\text{dom}}$, bypassing the index entirely; unindexed factual queries fall in buffer band $T_t < d^* \leq R_{\text{dom}}$, raising a *Semantic Fault*; and indexed queries ($d^* \leq T_t$) trigger an immediate Early-Exit, eliminating hallucinations at the source and conserving compute.

## 5.2 High-Dimensional Indexing Without Overcrowding

Indexing millions of facts in low dimensions ($D = 512$) leads to metric ball overlap. In production models ($D = 4096$ or 8192), concentration of measure on hyperspheres ensures random vectors exhibit near-perfect quasi-orthogonality ($\langle u, v \rangle \approx 0$).

For query throughput, querying registry $\mathcal{F}_{\text{mono}}$ avoids exhaustive $O(M)$ scans. Leveraging graph-based Approximate Nearest Neighbor structures (e.g., HNSW or IVF-PQ via GPU FAISS), query complexity is sub-linear $O(\log M)$ with latencies $< 15\,\mu$s, negligible relative to a 20–50 ms transformer forward pass.

# 6 Conclusion

Attempting to eliminate polysemanticity to force interpretability resembles dismantling associative cognition because it does not mirror a relational database.

With **HySA**, I propose an engineering paradigm grounded in real systems: permitting polysemantic units to perform fluid, associative reasoning (governed by $T_p$), complemented by a deterministic verification substrate anchored by monosemantic Inodes ($T_m$). Rather than two disconnected models in series or parallel, HySA formalizes a unified substrate querying internal metric APIs whenever fluid narrative must yield to factual precision.

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